

OpenDoor (Evacuee Housing Matcher)

Project Name & Elevator Pitch

OpenDoor is an emergency accommodation marketplace connecting displaced families with citizens offering spare rooms, empty apartments, or temporary hosting. Finding immediate, safe housing for evacuees is highly complex due to specific needs (large families, pets, disabled accessibility, proximity to hospitals). OpenDoor allows hosts to list available spaces with detailed tags and availability calendars, while displaced families can search, filter, and request to stay. It acts as a secure, verified bridge for community-driven housing relief.

Core User Roles

- **Host:** Lists available accommodations, manages availability calendars, and approves or denies stay requests.
- **Evacuee Family:** Searches for matching properties, requests bookings, and manages their stay details.
- **Verification Admin:** A platform moderator who verifies the identity of both hosts and evacuees to maintain safety.

User Stories

Role: Host

- As a Host, I want to create a property listing with specific tags (e.g., "Has Safe Room/Mamad", "Pet Friendly", "Wheelchair Accessible") so that families with specific needs can find it.
- As a Host, I want to define available date ranges for my property so that I am not requested during days I cannot host.
- As a Host, I want to receive and review booking requests from evacuees (including family size and origin city) so that I can approve or deny them.
- As a Host, I want my exact address to remain hidden from the public search, revealing it only to an evacuee *after* I approve their request.
- As a Host, I want to manually block out dates on my calendar if my availability changes so that I don't receive impossible requests.

Role: Evacuee Family

- As an Evacuee Family, I want to search for available housing by date range, required capacity, and specific safety tags so that my family's needs are met.

- As an Evacuee Family, I want to submit a booking request to a Host with a brief introductory message so that they understand our situation.
- As an Evacuee Family, I want to view a dashboard of my "Pending", "Approved", and "Rejected" housing requests so that I can manage our relocation plans.
- As an Evacuee Family, I want to cancel an approved stay if we find alternative housing so that the Host can open their home to someone else.

Role: Verification Admin

- As a Verification Admin, I want to view a queue of new user registrations to review their provided ID/documents.
- As a Verification Admin, I want to mark a user account as "Verified" or "Rejected" so that the platform remains secure and trusted.
- As a Verification Admin, I want to suspend a listing if it is reported as fraudulent or unsafe.

Out of Scope

- Payment processing or deposits (all exchanges are strictly volunteer/free)
- Legal lease generation or digital signatures
- Automated ID scraping/verification (admins will do this manually via uploaded photos)
- Complex multi-city routing
- Embedded interactive map interfaces (standard list/search views are sufficient)

Stretch / Research Zone

- **File Uploads (Security/Verification):** Allow users to upload an image of an ID or utility bill during registration for admin verification.
- **OAuth2 Authentication:** Allow users to sign in via Google to increase baseline trust and prevent spam account creation.

Team Split Suggestion

- **Backend Pair:** Build the complex search query logic (filtering by intersecting date ranges, tags, and capacity is non-trivial SQL). Implement role-based access control (RBAC) to ensure only approved evacuees can fetch exact addresses.
- **Frontend Pair:** Develop the calendar UI for hosts to easily block out dates. Build a multi-step registration flow with conditional rendering based on user type (Host vs. Evacuee).